

CMI Entrance Examination — Mock Test (Part A)

Instructions. For each of the following multiple-choice questions, one or more of the given options may be correct. You must identify all correct options to get full credit.

Q1.

Suppose you have three distinct, non-zero real numbers a , b , and c with $a + b + c \neq 0$. Which of the following expressions will always simplify to a quadratic polynomial (degree exactly 2) for any real number x ?

- A) $\frac{(x-b)(x-c)}{(a-b)(a-c)} + \frac{(x-c)(x-a)}{(b-c)(b-a)} + \frac{(x-a)(x-b)}{(c-a)(c-b)}$
- B) $\frac{x(x-b)(x-c)}{(a-b)(a-c)} + \frac{x(x-c)(x-a)}{(b-c)(b-a)} + \frac{x(x-a)(x-b)}{(c-a)(c-b)}$
- C) $\frac{bc(x-a)^2}{(a-b)(a-c)} + \frac{ca(x-b)^2}{(b-c)(b-a)} + \frac{ab(x-c)^2}{(c-a)(c-b)}$
- D) $\frac{a^2(x-b)(x-c)}{(a-b)(a-c)} + \frac{b^2(x-c)(x-a)}{(b-c)(b-a)} + \frac{c^2(x-a)(x-b)}{(c-a)(c-b)}$

Q2.

Consider an $n \times n$ matrix A with real entries that satisfies the property $A^2 = A$ (such matrices are called idempotent). No matter which specific matrix A you choose with this property, which of the following matrices is/are guaranteed to be invertible?

- A) $I + A$
- B) $I - A$
- C) $I + 2A$
- D) $I + \frac{1}{2}A$

Q3.

Consider the integral $I_n = \int_0^1 \frac{x^n}{1+x} dx$ defined for any integer $n \geq 0$. As you look at the sequence of values generated by I_n , which of these statements correctly describe its behavior?

- A) $\lim_{n \rightarrow \infty} I_n = 0$
- B) $I_n + I_{n+1} = \frac{1}{n+1}$ for all $n \geq 0$

C) $I_n < \frac{1}{2(n+1)}$ for all $n \geq 0$

D) $I_n > \frac{1}{2(n+1)}$ for all $n \geq 0$

Q4.

Let's say you have an infinitely smooth (infinitely differentiable) function $f: (-1, 1) \rightarrow \mathbb{R}$. Interestingly, this function hits zero at every point of the form $x = \frac{1}{n}$ for all integers $n \geq 2$. Based on this quirky property, which of the following statements absolutely must be true?

A) $f^{(k)}(0) = 0$ for all integers $k \geq 0$ (where $f^{(k)}$ denotes the k -th derivative).

B) $f(x) = 0$ identically for all $x \in (-1, 1)$.

C) The Taylor series of f centered at $x = 0$ converges to 0 for all $x \in (-1, 1)$.

D) There exists at least one real number $c \in (0, \frac{1}{2})$ such that $f''(c) = 0$.

Q5.

Consider a sequence $\{a_n\}$ formed by multiplying terms together, specifically

$$a_n = \prod_{k=1}^n \left(1 + \frac{k}{n^2}\right) \quad \text{for } n \geq 1.$$

How does this sequence behave as n gets larger? Pick the true statement(s).

A) The sequence $\{a_n\}$ converges to $\sqrt{e}$.

B) The sequence $\{a_n\}$ converges to e .

C) The sequence $\{a_n\}$ is strictly increasing.

D) $\lim_{n \rightarrow \infty} \ln(a_n) = \frac{1}{2}$

Q6.

Suppose $f(x)$ is a function defined on the non-negative real line $[0, \infty)$, and you are told it is uniformly continuous everywhere on this domain. Which of these properties will $f(x)$ always possess?

A) If the improper integral $\int_0^\infty f(x) dx$ converges to a finite value, then $\lim_{x \rightarrow \infty} f(x) = 0$.

B) There exist positive real constants A and B such that $|f(x)| \leq Ax + B$ for all $x \geq 0$.

C) $f(x)$ must be a bounded function on $[0, \infty)$.

D) If $f(x)$ is additionally known to be continuously differentiable on $(0, \infty)$, then its derivative $f'(x)$ must be bounded.

Q7.

Let p be a prime number, and let's explore the integer solutions (x, y) to the equation $x^2 + y^2 = p$. Which of the following statements about these solutions are correct?

- A) If $p \equiv 1 \pmod{4}$, the equation has exactly one solution in positive integers (x, y) up to the swapping of x and y .
- B) If $p \equiv 3 \pmod{4}$, the equation has no solutions in integers.
- C) The total number of integer solutions (x, y) (accounting for all signs and orderings) for $p = 13$ is 8.
- D) If (x, y) is an integer solution for some odd prime p , then the sum $x + y$ must be an odd integer.

Q8.

Imagine a finite set S that contains exactly n elements. If you choose an ordered pair of subsets (A, B) from S , which of the following counting formulas will give you the right answer?

- A) The number of pairs (A, B) such that $A \cup B = S$ is 3^n .
- B) The number of pairs (A, B) such that $A \subseteq B$ is 3^n .
- C) The number of pairs (A, B) such that $A \cap B = \emptyset$ is 3^n .
- D) The number of pairs (A, B) such that $|A| = |B|$ is $\binom{2n}{n}$.

Q9.

Picture a triangle ABC and its incenter I . If you draw a line straight through A and I , it will eventually hit the triangle's circumcircle again at a new point, D . What interesting geometric properties does this point D have?

- A) $DI = DB = DC$.
- B) The point D is the exact center of the circumcircle of $\triangle BIC$.
- C) The length $AI \cdot ID = R^2 - d^2$, where R is the circumradius of $\triangle ABC$ and d is the distance between the incenter and the circumcenter.
- D) The point D acts as the orthocenter for $\triangle BIC$.

Q10.

For any positive integer n , define the sum

$$S_n = \sum_{k=1}^n \tan^2\left(\frac{k\pi}{2n+1}\right).$$

If you think about these terms as the roots of a specific polynomial, which of the following identities perfectly evaluate the sum or product?

A) $S_n = n(2n + 1)$

B) The polynomial whose exact roots are $x_k = \tan^2\left(\frac{k\pi}{2n+1}\right)$ for $k = 1, 2, \dots, n$ is of degree n .

C) $\sum_{k=1}^n \sec^2\left(\frac{k\pi}{2n+1}\right) = 2n(n+1)$

D) $\prod_{k=1}^n \tan^2\left(\frac{k\pi}{2n+1}\right) = 2n+1$